

topology.tcl – creates the network topology

dc.tcl – creates the servers and VMs

setup_params.tcl – general configuration of servers, switches, tasks, etc

user.tcl – defines the users and their behaviour

record.tcl – reports the results

finish.tcl – prints the statistics

The output can be viewed via the browser using the *show-dashboard.html* file by running the *./run* script.

The *./run* script consists of the following parameters: data centre load, simulation time and memory requirement. The data centre load specifies the value from 0 to 1 (values near 0 indicate the idle data centre, while the load closer or greater than 1 indicates saturation of the data centre). The simulation time specifies the task that can be scheduled under a VM or a single host, based on the deadlines of the task.

The simulation results are processed in the *~greencloud/traces/* directory. There are various trace files that record the information from the data centre: load, main tasks, switch tracing, loading, etc.

Changing the parameters of the cloud

The parameters of the data centre can be changed using the Tcl files that were shown in the previous section. A simple change is shown below. Two files (*main.tcl* and *topology.tcl*) are modified catering to 40 servers and a single user cloud data centre with an average load capacity of 0.3 (as shown in Table 1).

```
#topology.tcl, where the network topology is been set
switch $sim(dc_type) {
    "three-tier high-speed" {
        set top(NCore) 2 ;# Number of L3 Switches in
the CORE network
        set top(NAggr) [expr 2*$top(NCore)] ;# Number of
Switches in AGGREGATION
        set top(NAccess) 256 ;# Number switches in
ACCESS network
        set top(NRackHosts) 3 ;# Number of Hosts on a
rack
    }
    "three-tier debug" {
        set top(NCore) 1 ;# Number of L3 Switches in
the CORE network
        set top(NAggr) [expr 2*$top(NCore)] ;# Number of
Switches in AGGREGATION
        set top(NAccess) 2 ;# Number switches in
ACCESS network per pod
        set top(NRackHosts) 20 ;# Number of Hosts on a
rack
    }
    three-tier
default {
    set top(NCore) 8 ;# Number of L3 Switches in
the CORE network
```

```
        set top(NAggr) [expr 2*$top(NCore)] ;# Number of
Switches in AGGREGATION
        set top(NAccess) 64 ;# Number switches in
ACCESS network
        set top(NRackHosts) 3 ;# Number of Hosts on a
rack
    }
}

# Number of racks is set as 2 * 1
set top(NRacks) [expr $top(NAccess)*$top(NCore)]

# Number of servers is set to 2 * 20 (40 servers)
set top(NServers) [expr $top(NRacks)*$top(NRackHosts)]
....

#main.tcl, where the simulation information and data centre
load information is specified
# Type of DC architecture
set sim(dc_type) "three-tier debug"

# Set the time of simulation end
set sim(end_time) [ expr 60.1 + [lindex $argv 1] ] ;#
simulation length set to 60 s + deadline of tasks

# Start collecting statistics
set sim(start_time) 0.1

set sim(tot_time) [expr $sim(end_time) - $sim(start_time)]

set sim(linkload_stats) "enabled"

# Set the interval time (in seconds) to make graphs and to
create flowmonitor file
set sim(interval) 0.1

# Setting up main simulation parameters
source "setup_params.tcl"

# Get new instance of simulator
set ns [new Simulator]

# Tracing general files (*.nam & *.tr)
set nf [open "../traces/main.nam" w]
set trace [open "../traces/main.tr" w]

# Building data centre topology
source "topology.tcl"
.....
```

The graph in the browser shows four parts: the simulation data as shown in Table 1, the data centre characteristics as shown in Figure 2, the DC network characteristics as shown in Figure 3 and the energy consumption details as shown in Figure 4.